Giacomo Cappelletto

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Technical Profile

Computer Engineering student at Boston University with a 3.97 GPA, building production software, LLM-agent workflows, data systems, applied ML pipelines, and robotics autonomy systems. Strongest in Python, TypeScript, React/Next.js, SQL/PostgreSQL, Rust, Databricks, Flask/Dash, and cloud-deployed AI applications.

Education

- **Boston University, Boston, MA** — 2024 – 2028
  B.Sc. in Computer Engineering; GPA: 3.97
- **H-Farm International School, Venice, Italy** — 2022 – 2024
  International Baccalaureate (IB) Diploma — *HL: Mathematics, Physics, Computer Science*

Technical Experience

- **Boston University, Prof. Brian Kulis, Boston, MA** — Aug 2026 – Present
  Undergraduate Machine Learning Researcher
  - **Pose-Sequence Retrieval**: Researching robust contextual metric learning for human-motion retrieval, learning fixed-length embeddings from temporal 2D/3D pose sequences for nearest-neighbor search across labeled movement classes.
  - **Contextual Metric Learning**: Adapting supervised contextual-similarity optimization from image retrieval to motion embeddings, comparing contextual + contrastive objectives against triplet, supervised contrastive, and classification-based retrieval baselines.
  - **Robustness Evaluation**: Designing controlled experiments for pose-estimation noise, missing joints, temporal jitter, occlusion, and viewpoint shift, evaluated with Recall@K and mAP on public skeleton/motion benchmarks.
- **Banca Mediolanum, Milan, Italy** — Jun 2026 – Present
  Software / AI Intern, Customer Knowledge
  - **Technical Ownership**: Owned an internal AI-agent runtime, enterprise chat app, and prompt-governance platform across architecture, full-stack engineering, evaluation, deployment, and stakeholder iteration.
  - **Agent Platform Engineering**: Built a Databricks Model Serving agent with MLflow ResponsesAgent and LangGraph ReAct; designed a versioned Pydantic contract, policy-controlled tool execution, request-scoped context budgets, governed dynamic SQL, citation-backed document RAG, evidence-grounded review, and MLflow tracing.
  - **AI Product Engineering**: Built a React/Vite + FastAPI Databricks App with SSE streaming, authenticated sessions, structured renderables, Lakebase/PostgreSQL, MLflow feedback, and Asset Bundle deployment.
  - **Evaluation & Governance**: Built a Dash/Flask platform for append-only Delta prompt versioning, permission-aware workflows, side-by-side benchmarks, LLM judging, MLflow GenAI scoring, and Unity Catalog quality controls.
- **Boston University College of Engineering, Boston, MA** — Jan 2026 – May 2026
  Undergraduate Machine Learning Researcher
  - **Computer Vision Pipeline**: Built a Python biomechanics pipeline to stabilize rowing video, run single-athlete 2D keypoint inference with MMPose, lift poses to 3D skeletons with MotionBERT, and output per-frame joint positions and angles.
  - **Kinematic Analysis**: Computed stroke-averaged 3D kinematics to quantify coordination, range of motion, technical consistency, and timing differences across rowing strokes.
  - **Sequence Modeling**: Trained sequence-to-sequence regression models mapping time-aligned 3D kinematics to rowing force curves, validated against instrumented on-water and erg telemetry.
- **Boston University Men's Rowing, Boston, MA** — Jan 2026 – May 2026
  Software Engineer
  - **Team Training Platform**: Built a mobile-first training tracker for 50 BU Men's Rowing athletes and coaches, unifying athlete logging, coach review, proof-validated entries, leaderboards, configurable weekly cutoffs, and team-specific requirements.
  - **Product Iteration**: Iterated directly with athletes and coaches to add requested features and improve team workflows, with expansion planned to two additional teams for a projected 150 total users.

- **Backend + Automation**: Implemented Supabase and Prisma-backed data models with role-based access, proof retention, scalable reporting, and automated weekly recap emails for athlete and coach workflows.

- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2025 – Oct 2025
  Software Engineer, Tauri Inventory Management System Rebuild
  - **Migration + Performance**: Rebuilt an Electron inventory app using Tauri, React, and Rust, reducing app size by 70%, memory usage by 80%, and startup time by 60%; improved Shopify inventory search by 43% through parallel API processing and GraphQL-optimized queries.
  - **Production Features**: Built SKU-based instant search, Firebase-backed audit history, real-time inventory reconciliation, multi-location update workflows, GitHub Releases distribution, and auto-update deployment.
- **TickIT GmbH, Remote** Nov 2024 – Feb 2025
  Frontend & Backend Engineer
  - **Collaborative Platform Development**: Contributed roughly 30,000 lines of code across a 150,000-line online event-ticketing platform as part of a four-developer remote team, working across Ruby backend services and Next.js/TypeScript frontend interfaces.
  - **Ticketing + Access Control**: Developed QR-code ticket generation and validation infrastructure, ticket purchasing flows, event pages, ticket validation interfaces, and friend-sharing features for attendee access management.
  - **Analytics + Forecasting**: Built internal customer spending tracking systems, analytics pipelines, and organizer dashboards for spending predictions, inventory forecasting, operational metrics, and event cost planning; platform payments were processed through Stripe.
  - **Engineering Collaboration**: Participated in remote feature planning, cross-stack implementation, and code reviews, contributing central functionality for event access control and organizer analytics.

Selected Projects

- **NoteWorthy – Automated Note Conversion Platform**: Built noteworthy.vercel.app, a Next.js/TypeScript application that converts handwritten notes into styled PDFs and LaTeX. Integrated Google AI Studio, GitHub/Google OAuth, and a Dockerized LaTeX compiler; deployed through Google Cloud Run and Vercel.
- **CIFAR-10 Image Recognition with Neural Architecture Search**: Built a TensorFlow/Keras CIFAR-10 classifier and neural architecture search pipeline to generate, train, evaluate, and select CNN candidates with automated metric logging and best-model selection.
- **Deskinator – Autonomous Desk-Cleaning Robot**: Built Python robotics software for a Raspberry Pi desk-cleaning robot using APDS9960 edge sensing, stepper odometry, RANSAC rectangle fitting, boustrophedon coverage planning, telemetry, and simulation-based performance analysis.

Leadership & Athletics

- **Boston University Men's Rowing, Boston, MA** Sep 2024 – Present
  NCAA Division I Student-Athlete
  - **Performance + Time Management**: Compete at the Division I and national level while balancing 20+ hours per week of training, competition, travel, and team commitments with a full Computer Engineering course load.
  - **Awards**: Freshman Student-Athlete of the Year (2025); Most Improved Oarsman (2026).

Skills

- **Languages**: Python, TypeScript/JavaScript, SQL, Rust, Ruby, C/C++, Swift, Java
  AI/ML: metric learning, contextual similarity, pose embeddings, LLM agents, RAG, semantic search, MLflow GenAI Evaluation, TensorFlow/Keras, MMPose, MotionBERT, RANSAC, Ollama
  Frameworks: Next.js, React, Node.js, Flask/Dash, Ruby on Rails, Streamlit, Tauri, SwiftUI, Tailwind CSS, shadcn/ui
  Data/Cloud: Databricks Apps, Unity Catalog, Delta tables, Vector Search, PostgreSQL, Supabase, Prisma, Firebase, Docker, Google Cloud Run, Vercel
  Tools: Git/GitHub, Raspberry Pi, Shopify GraphQL API, Stripe, GitHub Releases, AI-assisted development workflows